UNIT – V : CAPP, Production Planning & Trends

1. Computer Aided Process Planning (CAPP)

- **Process planning** = deciding **how** a part will be made: selection of **processes, machines, tools, fixtures, cutting data, operation sequence and times** – result is the **route sheet / operation sheet** (process plan). It links **design and manufacturing**.
- Manual process planning depends on the **experience of the planner** → inconsistent, slow, skilled labour needed.
- **CAPP** = use of the computer to prepare the process plan automatically (or semi-automatically).

Steps in process planning

- Study the drawing → select raw material → select processes and sequence → select machines → select tools, fixtures, gauges → fix cutting conditions (speed, feed, depth) → estimate time & cost → prepare the route sheet / NC program.

Benefits of CAPP

- Standardised and **consistent** plans, **faster planning**, less skill needed; reduces planning staff and lead time.
- Better use of the shop's resources, lower manufacturing cost, links with **CAD / CAM / MRP / NC programming**, easy updating of plans, supports GT and FMS, possibility of optimum plans.

2. Retrieval Type CAPP (Variant approach)

- Based on **Group Technology**: parts are classified and coded into **part families**; a **standard process plan (master plan)** is stored for each family.
- Procedure: code the new part → find its family → **retrieve the standard plan** → **edit / modify** it for the particular part → print the route sheet.

RETRIEVAL (VARIANT) CAPP



standard plans stored in database → retrieved & modified → new plan

- Two stages: **preparatory stage** (building the database: coding, grouping, creating standard plans) and **production stage** (using the system).
- Popular systems: **CAPP (CAM-I)**, MIPLAN, MultiCAPP, CPPP.

Advantages

- Simple, quick to develop, low cost, easy to learn; the planner keeps control; good where parts are similar and a large number of families exist.

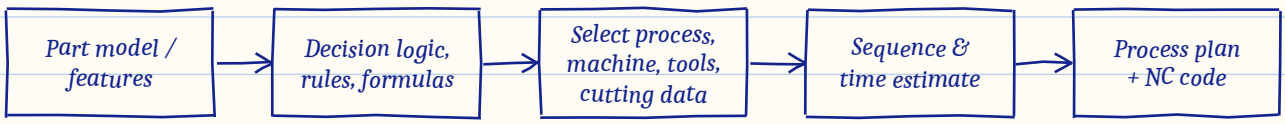
Disadvantages

- Needs a **GT coding** effort; plans are only for existing families; new / dissimilar parts need manual planning; quality depends on standard plans and the planner who edits; cannot optimise.

3. Generative Type CAPP

- Process plan is **generated automatically** from scratch for each new part using **decision logic (rules), formulas, technology data and algorithms** – no standard plan is retrieved.
- Input: complete description of the part – a **CAD model with features** (holes, slots, steps, tolerances, surface finish, material) – plus manufacturing knowledge in the computer.

GENERATIVE CAPP



plan is created fresh by the computer using logic – no standard plan needed

- **Knowledge base**: machine capabilities, tooling database, machinability data, cost / time formulas and the **decision tables / decision trees / expert-system rules** (IF feature = hole of Ø10, depth 30 THEN drill with a 10 mm drill).

- Techniques: decision trees & tables, **expert systems (AI)**, feature recognition from CAD, **process-capability matching**.
- Popular systems: AUTAP, **CIMS/PRO**, XPLAN, **GARI**, Part-Tec.

Advantages

- Fully automatic, **consistent and optimised** plans; handles new parts; direct link to CAD and NC programming; fast for complex parts; no GT coding needed.

Disadvantages

- Very difficult and costly to develop (process knowledge must be converted to logic); needs **feature-based CAD**; maintenance of rules; **semi-generative** (hybrid) systems combine both approaches in practice.

Retrieval vs generative

Feature	Retrieval (variant)	Generative
Basis	standard plan of a part family	logic / rules and the CAD model
GT code needed	yes	not necessary
New part	needs modification by the planner	plan generated automatically
Development	easy, cheap	difficult, expensive
Optimisation	limited	possible
Use	repetitive, similar parts	complex, varied parts; CAD / CAM integration

4. Machinability Data Systems

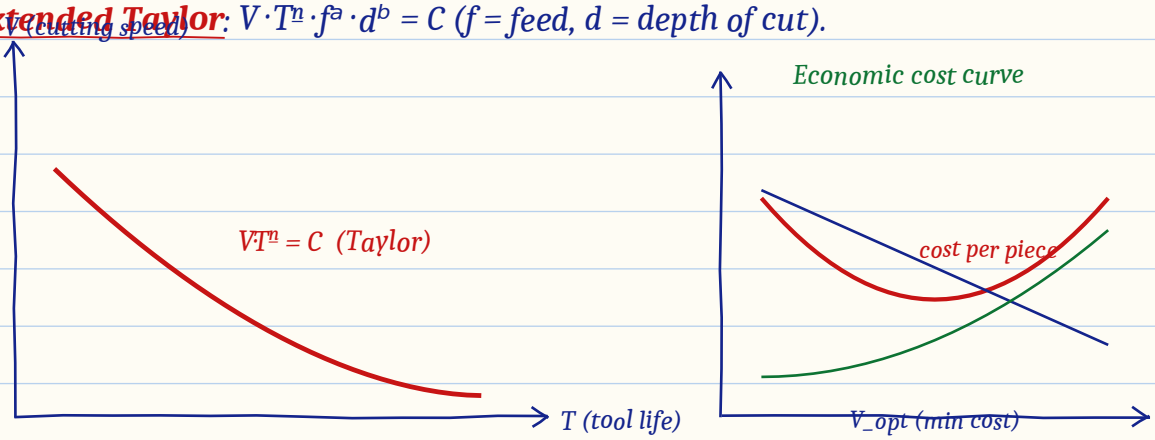
- **Machinability** = ease with which a material can be machined (tool life, surface finish, cutting force, power, chip form). Machinability data = recommended **cutting speed, feed, depth of cut, tool material and geometry, coolant** for a given work material and operation.
- **Machinability data system** = computerised database + models that supply the optimum cutting conditions to the process planner / NC programmer.
- Two forms: **database (look-up tables)** from handbooks and experiments, and **mathematical models** (Taylor's equation) calculating the optimum values.

$$V \cdot T^n = C \quad \text{(Taylor's tool life equation)}$$

- V = cutting speed (m/min), T = tool life (min), n = exponent (HSS 0.1 to 0.15, carbide

0.2 to 0.5, ceramic 0.4 to 0.7), C = constant.

- **Extended Taylor**: $V \cdot T^n \cdot f^a \cdot d^b = C$ (f = feed, d = depth of cut).



- **Optimum cutting speed** may be selected for (a) **minimum cost** per piece, (b) **maximum production rate** (minimum time), or (c) **maximum profit**. Min-cost speed is lower than max-production-rate speed.

$$T(\text{min cost}) = (1/n - 1)(Tc + C_t/C_m)$$

- Tc = tool change time, C_t = tool cost per cutting edge, C_m = machine + labour cost per minute.

$$T(\text{max production}) = (1/n - 1) \cdot Tc$$

- **Example**: for carbide n = 0.25, Tc = 2 min: T(max production) = (4 – 1) × 2 = 6 min.
- Data are obtained from experiments, handbooks (Machining Data Handbook), manufacturers; examples of systems: **MACHDATA**, **METCUT**, **Machinability Data Center (MDC)**, **Kennametal Novo**.
- Used in CAPP, NC programming and adaptive control; **advantages**: optimal cutting conditions, less trial and error, longer tool life, lower cost, shorter cycle time.

5. Computer Generated Time Standards

- **Time standard** = the time allowed to a qualified, trained worker working at normal pace to complete one operation (or job) – used for **scheduling, costing, wage incentives, capacity planning, plant layout**.
- Conventional method: stop-watch time study and work sampling – needs the job to be present and an observer.
- **Computer-generated time standards** (standard data systems / predetermined time systems): a computer calculates the **standard time from the process plan** before the

part is made – no stop-watch.

Methods

- **Standard data method** – stored time values for elements (e.g. rapid traverse, load / unload, tool change) and **formulae for machining time** with cutting data.
- **Predetermined motion time systems (PMTS)** – MTM (Methods-Time Measurement), **MOST**, work factor: basic human motions are given fixed time values and summed.
- Programs: **MTM-based software**, **COMTEK**, **MTM-LINK**, **AUTOPLAN / CAPP time module**.

Computation of the standard time

- **Time per piece Tc = Tsu / Q + Tm + Th + Tt** → setup time per batch (Tsu / Q), machining time Tm, work handling time Th, tool handling time Tt.

$$Tm\ (turning) = L / (f \cdot N)\ \ min, \quad where\ N = 1000\ V / (\pi\ D)\ \ rpm$$

- **Standard time = Normal time × (1 + allowances)** – allowances for personal needs, fatigue and delay (usually 10 to 20%).
- **Example**: Turning a Ø50 mm bar, length 120 mm, V = 100 m/min, f = 0.2 mm/rev: N = 1000 × 100 / (π × 50) = 637 rpm; Tm = 120 / (0.2 × 637) = **0.94 min**; with handling 0.6 min + allowance 15% → standard time = (0.94 + 0.6) × 1.15 = **1.77 min / piece**.

Advantages

- Fast and consistent, available at the planning stage, no disturbance to the workers, easy updating when methods change, integrates with CAPP, MRP and costing.

Computer Integrated Production Planning

6. Production Planning and Control (PPC) – Overview

- **PPC** = planning, scheduling and controlling of production so that the right product is made in the **right quantity, at the right time, at the lowest cost**.
- Functions: forecasting, **aggregate planning**, **master production schedule (MPS)**, material requirements planning, **capacity planning**, scheduling & dispatching, shop-floor control, inventory control, purchasing, and cost control.
- **Computer integrated production planning (CIPP)** = these functions performed by computer systems sharing a common database; information from CAD (BOM), CAPP (routings, times) and the shop floor (status) feed the planning.

7. Capacity Planning

- **Capacity** = maximum output a facility (machine, work centre, plant) can produce per period (hours / week). **Capacity planning** = determine the labour and machine capacity needed to meet the production plan and balance it against the available capacity.
- Levels: **Resource requirements planning (long range)**, **Rough-cut capacity planning (RCCP – from MPS)** and **Capacity requirements planning (CRP – from MRP, medium / short range)**.

Steps

- Find the **load** (required hours) on each work centre from the planned orders and routings (set-up + run time) → compare with **available capacity** → find overload / underload → adjust.

Required hours = Σ (Tsu + Q × Tp) of all orders on the work centre

Utilisation = (hours used) / (hours available); Available capacity = machines × hours × efficiency × utilisation

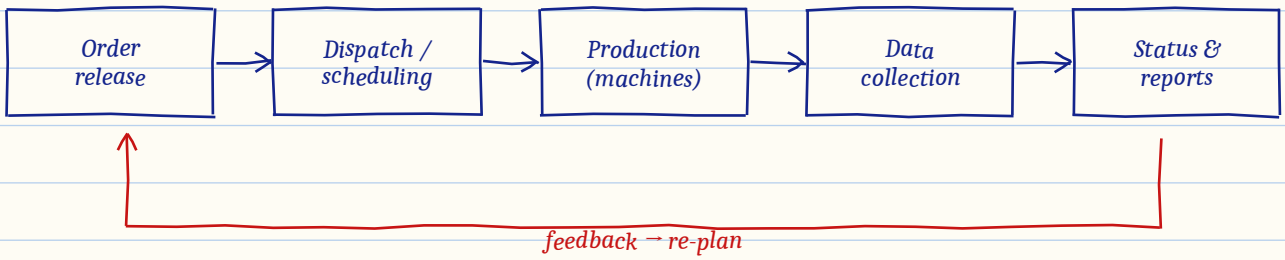
- **Example**: a work centre has 2 machines × 40 h/week × 85% efficiency = 68 h available. Orders need 80 h → overload 12 h.
- Ways to correct: overtime, extra shift, subcontracting, alternate machines / routing, rescheduling (level loading), add machines, hire labour. If capacity is too

high → **underload** → reduce shifts / take more orders.

- **Finite loading** (do not exceed capacity) vs **infinite loading** (load without limits, then adjust). Capacity strategies: lead, lag, match.
- **CAPP + MRP** supply the data (routing and standard times) for CRP – shows the use of computer integration.

8. Shop Floor Control (SFC)

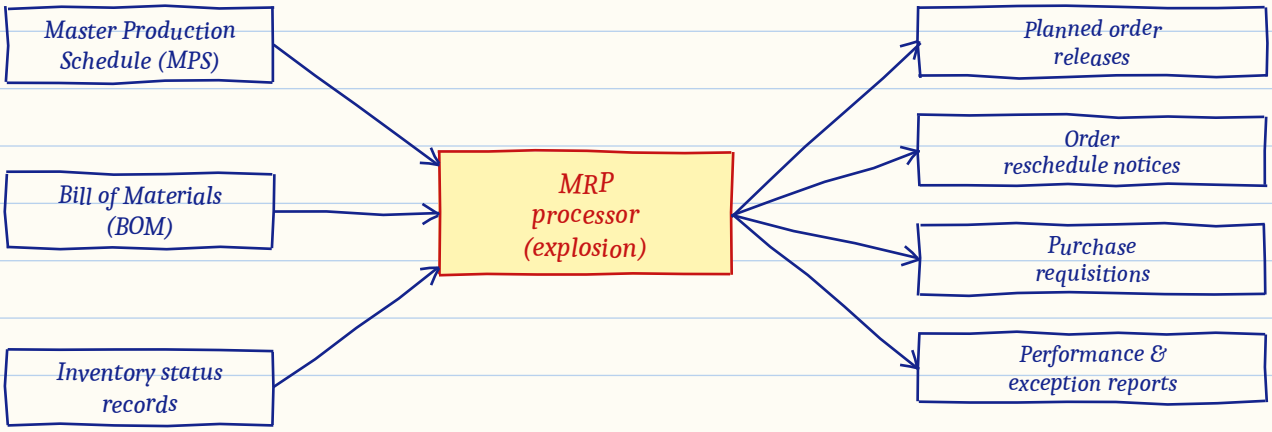
- **SFC** = system for **releasing orders to the shop, scheduling and dispatching them, and monitoring progress** on the shop floor; it follows after MRP. Provides the **feedback** to planning.
- Three phases: **(1) Order release** – authorise the production order with the documents (route sheet, drawings, material, tooling); **(2) Order scheduling and dispatching** – sequence the jobs at each work centre (priority rules like **FCFS, SPT, EDD, critical ratio**); **(3) Order progress** – monitor status, work-in-process, queue, exceptions.



- **Data collection (shop-floor data collection, SFDC)**: bar-code, RFID, **touch screens, machine monitoring, counters** – operators report start / stop, quantity, scrap; automated collection gives real-time status.
- Reports: work order status, **work-in-process**, **exception reports (late jobs)**, machine utilisation, scrap, **labour efficiency**.
- **Input / output control** and **priority control**; **MES (Manufacturing Execution System)** and **Kanban** are the modern forms.
- **Factory data collection systems** are the link of CIM between the shop floor and the planning system.

9. MRP – Material Requirements Planning (MRP-I)

- **MRP** = computer-based system that determines **what materials (components, sub-assemblies, raw materials) are needed, how many and when**, based on the master production schedule (MPS), the bill of materials and the inventory data.
- Applies to **dependent-demand items** (parts needed for end products), unlike reorder-point systems used for independent demand.
- **Inputs**: (1) **Master production schedule (MPS)** – what end items and when; (2) **Bill of materials (BOM)** – product structure (levels, quantity per parent); (3) **Inventory status file** – on-hand, on-order, lead times, lot-size rules, safety stock.



- **Outputs**: planned order releases, order reschedule notices (expedite / de-expedite), purchase requisitions, exception and performance reports.

MRP logic – the calculation (explosion)

- **Gross requirements** (from MPS or parent) – **scheduled receipts** – **projected on-hand** = **net requirements**.
- Net requirements are converted to **planned order receipts** using a **lot-sizing rule** (lot-for-lot, EOQ, fixed order quantity, period order quantity).
- **Planned order release** = planned receipt **offset by lead time**.

Net requirement = Gross requirement – (On hand + Scheduled receipts) + Safety stock

- Done level by level (low-level coding) from the end item down to the raw material – **BOM explosion**.

Example – MRP table (lot-for-lot, lead time 2 weeks, on hand 20)

Week	1	2	3	4	5	6
Gross requirement	0	0	40	0	60	30
On hand (end)	20	20	0	0	0	0
Net requirement	0	0	20	0	60	30
Planned order receipt	0	0	20	0	60	30
Planned order release	20	0	60	30	0	0

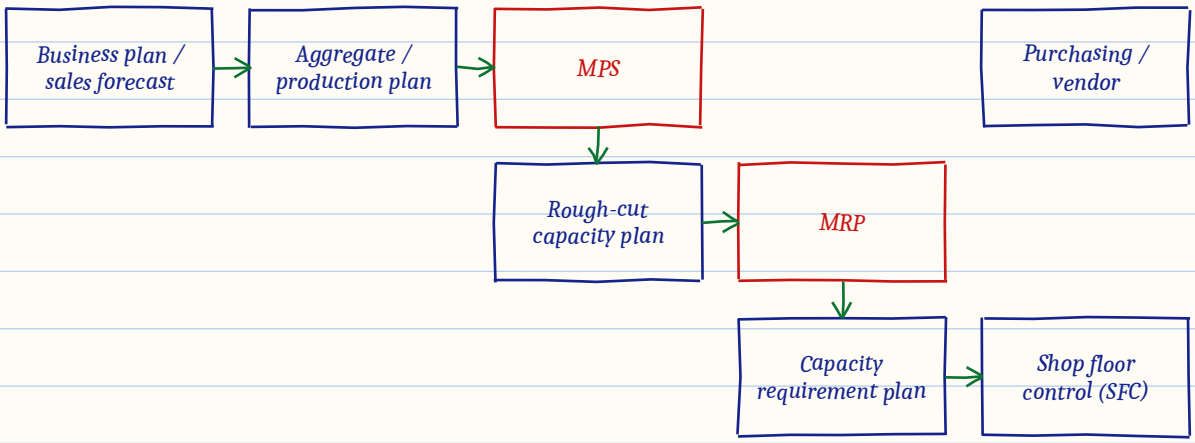
- Week 3: 40 – 20 on hand = 20 net → receipt in week 3, so the order is **released 2 weeks earlier (week 1)**; the 60 needed in week 5 is released in week 3, and the 30 in week 6 in week 4.

Benefits and limitations of MRP-I

- Benefits: lower inventory, fewer shortages, better delivery performance, priorities for the shop, planning of purchases.
- Limitations: **assumes infinite capacity** and fixed lead times; no feedback on capacity; accuracy of data (BOM, inventory) is vital; only materials planning.

10. MRP-II – Manufacturing Resource Planning

- **MRP II** = extension of MRP-I into a **closed-loop** system that plans **all the resources** of the company – materials, **capacity (machines, labour), finance, marketing** – and links with **simulation ("what-if")**. (Oliver Wight, 1980s.)
- Closed-loop MRP: MPS → MRP → **capacity requirements planning** → **shop floor control / purchasing** → **feedback** to the planning levels.



feedback (closed loop) from SFC to planning levels

Modules of MRP II

- **Business planning, sales and operations planning (aggregate planning)**, demand management & forecasting, **MPS**, **rough-cut capacity planning**, **MRP**, **capacity requirements planning**, **shop-floor control**, purchasing, inventory management, **financial and cost accounting**, distribution resource planning (DRP), performance measurement.

MRP-I vs MRP-II

MRP-I	MRP-II
Material planning only	Plans all the resources (material, capacity, money)
Open loop, infinite capacity assumed	Closed loop with capacity check and feedback
Output: order releases	Output: integrated plan with finance and marketing
Few modules, 1970s	Many modules; basis of ERP (1990s)

- **ERP** (Enterprise Resource Planning) extends MRP II to the whole enterprise: HR, finance, supply chain, CRM, etc. (SAP, Oracle).

11. CIMS – Computer Integrated Manufacturing Systems and their Benefits

- **CIMS** = the integrated system of **CAD, CAPP, CAM / CNC, FMS, CAQC, AS/RS, robots, MRP / PPC and business functions** connected by a **network and common database (LAN, MAP / TOP protocols)** – the "factory of the future".
- Levels (hierarchy): **plant / enterprise level → shop / area level → cell level → workstation level → equipment level**.

Benefits of CIMS

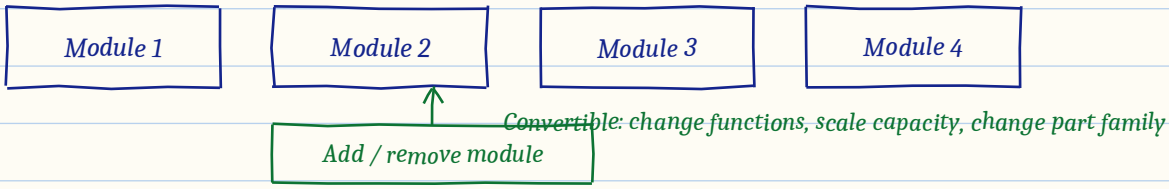
- **Shorter lead time**: 20 to 60% shorter design and production cycle; lower **inventory (WIP)** by 30 to 60%.
- Higher **productivity (40 to 70%)**, better utilisation of machines, less scrap and **better quality**, lower cost of products.
- **Flexibility** and quick response to customer demands; better **customer service**, low cost of changes.
- Better **management control** with real-time information, better decisions, lower indirect labour, no duplication of data.
- Competitive advantage in the global market. Disadvantages: very high investment, complexity, need for skilled people, compatibility of vendors' systems, resistance to change.

Trends in Manufacturing Systems

12. Reconfigurable Manufacturing

- **Reconfigurable manufacturing system (RMS)** (Y. Koren, 1999) = a system designed at the outset for **rapid change in structure** (hardware and software) so that it can quickly **adjust production capacity and functionality** in response to sudden market changes or product changes.
- It sits between dedicated lines (high volume, no flexibility) and FMS / CNC (high flexibility, lower throughput): RMS gives **customised flexibility** – only the flexibility needed for a part family.

Reconfigurable manufacturing system (RMS)



Six core characteristics of RMS

- **Modularity** – components (machines, spindles, fixtures, software) are modules that can be added / removed.
- **Integrability** – quick and precise integration of modules through standard interfaces.
- **Customisation** – flexibility limited to a part family (customised flexibility and control).
- **Convertibility** – ease of changing the functionality to new products (change spindle heads, tools).
- **Scalability** – capacity can be increased / decreased by adding or removing modules or stations.
- **Diagnosability** – quick identification of quality and reliability problems at the source.

RMS vs DML vs FMS

Feature	Dedicated (DML)	FMS	RMS
Flexibility	none	general	customised (part family)
Capacity change	fixed	fixed	scalable
Cost / throughput	low cost, high throughput	high cost, low throughput	moderate cost and throughput
Structure	fixed	fixed	changeable

- Benefits: faster launch of new products, lower response time to market changes, capacity matching demand, lower investment, better use of existing equipment (reuse of modules).

13. Sustainable Manufacturing

- **Sustainable manufacturing** = creation of manufactured products using processes that are **non-polluting, conserve energy and natural resources, and are economically sound and safe** for employees, communities and consumers (US Dept. of Commerce). Triple bottom line: **people – planet – profit**.
- Principles (**6R**): **Reduce, Reuse, Recycle, Recover, Redesign, Remanufacture**.
- Practices:
  - **Green design / design for environment (DfE)**: material selection (recyclable, non-toxic), design for disassembly, durable and repairable products, lightweight.
  - **Energy efficiency**: efficient motors, energy monitoring, regenerative drives, renewable energy, heat recovery.
  - **Resource efficiency**: near-net-shape processes (casting, **additive manufacturing**), less scrap, **MQL / dry machining** (minimum quantity lubrication replaces flood coolant), water recycling.
  - **Waste management**: zero-waste to landfill, **end-of-life** recycling and remanufacturing, **closed-loop supply chain** and **circular economy**.
  - **Life-cycle assessment (LCA)**: environmental impact from raw material → manufacturing → use → disposal; carbon footprint; ISO 14001 environmental management.
- Enablers: Industry 4.0 / IoT sensors for energy monitoring, digital twins, simulation

to cut trial-and-error.

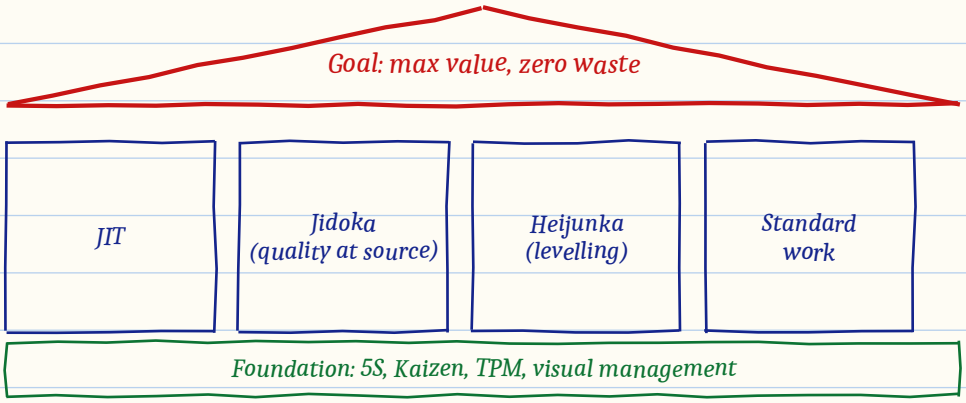
- Benefits: lower cost of energy and material, compliance with regulations, better brand image, safe working environment, long-term competitiveness. Challenges: initial cost, lack of data and standards, change of mind-set, supplier cooperation.

14. Lean Manufacturing

- **Lean manufacturing** (from the **Toyota Production System**, Taiichi Ohno) = a systematic approach to **eliminate waste (muda)** and create **maximum value for the customer** using the least resources.
- **Five lean principles** (Womack & Jones): (1) specify **value** from the customer's view, (2) identify the **value stream**, (3) create **flow**, (4) **pull** production, (5) pursue **perfection** (continuous improvement).

Seven wastes (TIMWOOD)

- **T**ransportation, **I**nventory, **M**otion, **W**aiting, **O**verproduction (worst), **O**ver-processing, **D**efects (+ 8th: unused creativity of people).



Lean tools and techniques

- **JIT (Just-In-Time)**: produce only what is needed, when needed, in the amount needed; **Kanban** (card signal system for pull); **Heijunka** (production levelling); **Takt time** = available time / customer demand.
- **Jidoka** (autonomation): machine stops automatically on a defect; **Poka-yoke** (mistake-proofing); **Andon** (alert system).
- **5S**: Sort (Seiri), Set in order (Seiton), Shine (Seiso), Standardise (Seiketsu), Sustain (Shitsuke).
- **Kaizen** (continuous improvement), **SMED** (single-minute exchange of die – quick changeover), **TPM** (total productive maintenance), **VSM** (value stream mapping), **cellular manufacturing / U-cells**, standard work, visual management.

- **Benefits**: lower inventory and lead time, better quality, less floor space, higher productivity, flexibility, employee involvement, lower cost.
- **Limitations / challenges**: needs stable demand and reliable suppliers, cultural change and training, vulnerable to disruptions (low buffers).

Lean, agile, sustainable – a comparison

Aspect	Lean	Reconfigurable	Sustainable
Focus	waste elimination, flow	quick change of capacity & function	environment, energy, resources
Tool	JIT, kaizen, 5S	modular machines	6R, LCA, green design
Result	low cost, high quality	responsiveness to market	long-term eco-efficiency

15. Quick Revision

CAPP: retrieval (variant, GT-based, edit standard plan) vs generative (logic, builds plan; CAD features).

- Machinability data: Taylor $VT^n = C$; min-cost vs max-production speed. Time standards: setup + machining + handling + allowance.
- Capacity planning: load vs available capacity; SFC: release – dispatch – monitor.
- MRP-I: MPS + BOM + inventory → planned orders; MRP-II: closed loop with capacity & finance; CIMS benefits.
- Trends: Reconfigurable (modularity, scalability, convertibility, integrability, customisation, diagnosability); Sustainable (6R, LCA); Lean (waste, JIT, kaizen, 5S, kanban).

Likely exam questions

- Explain retrieval and generative CAPP with block diagrams; compare them.
- What is a machinability data system? Explain Taylor's equation and its use.
- Explain computer generated time standards.
- Explain capacity planning and shop floor control.
- Explain MRP-I with inputs and outputs; differentiate between MRP-I and MRP-II.
- Write the benefits of CIMS.
- Explain reconfigurable, sustainable and lean manufacturing concepts.